

House Price Prediction – Project Summary

Introduction

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features. The dataset used for this project was the Housing Prices Dataset, which contains information such as area, number of bedrooms, bathrooms, stories, parking spaces, and additional amenities.

Data Preparation

The dataset was loaded and explored using Pandas. Missing values and duplicate records were checked and handled appropriately. Categorical features such as main road access, guest room availability, basement, air conditioning, and furnishing status were converted into numerical form using **one-hot encoding**. This preprocessing step ensured that the dataset was suitable for machine learning algorithms.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results and Findings

The Linear Regression model achieved an R^2 score of approximately **65.29%**, while the Random Forest Regressor achieved an R^2 score of approximately **61.19%**. Based on these results, Linear Regression performed slightly better on this dataset.

The analysis revealed that features such as area, number of bathrooms, number of stories, and air conditioning had a significant impact on house prices. The correlation heatmap and scatter plots also indicated a strong positive relationship between property size and price.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. The developed models were able to explain a substantial portion of the variation in house prices. Although the prediction accuracy was reasonable, additional features such as location details, neighborhood quality, and market conditions could further improve model performance. Based on the findings, real estate businesses should focus on property size and key amenities when estimating property values and making investment decisions.

Notes: Usually Random Forest performs better, but here give data set is very small ,also the relationship between features and prices is fairly linear